
NAAN MUDHALVAN

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team ID: SWTID-2026-2362

Team size: 5

Team Leader Name: Jarshika G

Team Members: Beula R, Varsha V, Siva Jeevitha J, Avanthika Eswari M

Women's Christian College, Nagercoil

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1. Introduction

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team Members

Team size: 5

Name	Role	Responsibility
Jarshika G	Team Leader	Project coordination, flow design and final documentation
Beula R	ServiceNow Developer	Standard Laptop catalog item and variables
Varsha V	Flow Designer Developer	Flow logic, approvals and catalog task creation
Siva Jeevitha J	Configuration Specialist	Users, groups, roles, ACLs and email notifications
Avanthika Eswari M	Tester	Test cases, validation and defect reporting

2. Project Overview

Purpose

In many organizations, requesting a standard laptop involves emails, spreadsheets and manual follow-ups between employees, managers, the procurement team and IT support. This causes delays, lost requests and no visibility into request status.

This project automates the entire laptop request lifecycle on the ServiceNow platform. An employee raises a request from the Service Catalog, and a flow built in Flow Designer handles approval, task creation, notifications and closure without manual intervention.

Goals

- Reduce the time taken to procure a standard laptop.
- Remove manual hand-offs by automating approvals and task assignment.
- Give employees and managers real-time visibility of request status.
- Maintain a consistent, auditable procurement process.

Key Features

- **Standard Laptop catalog item:** a single, easy-to-use request form with predefined variables.
- **Automated manager approval:** the flow routes the request to the requester's manager automatically.
- **Automatic task creation:** catalog tasks are assigned to the Procurement and IT Support groups.
- **Conditional logic:** the flow handles approved and rejected outcomes differently.
- **Email notifications:** requester and stakeholders are informed at every stage.
- **Low-code implementation:** built entirely with Flow Designer, with no custom scripting required.
- **Full audit trail:** every flow execution is recorded and can be inspected for troubleshooting.

3. Architecture

The solution is built on the ServiceNow platform and follows a three-layer model that maps to the frontend, backend and database of a traditional application.

Frontend: Service Catalog / Service Portal

Employees interact with the solution through the Service Portal (or the Employee Center). The Standard Laptop catalog item captures the request details through a short form.

Variables: Requested For, Business Justification, Needed By, Delivery Location and Accessories Required.

Backend: Flow Designer

Flow Designer is the automation engine. The flow is triggered when a requested item for the Standard Laptop catalog item is created, and it runs through the stages below.

Step	Flow Designer Element	Description
1	Trigger: Service Catalog	Starts the flow when a Standard Laptop request is submitted.
2	Action: Get Catalog Variables	Reads the values entered in the request form.
3	Action: Ask For Approval	Sends an approval to the requester's manager.
4	Flow Logic: If (Approved)	Branches the flow based on the approval decision.
5	Action: Create Catalog Task	Creates a procurement task for the Procurement group.
6	Action: Create Catalog Task	Creates a set-up and delivery task for the IT Support group.
7	Action: Update Record	Moves the requested item to Closed Complete once tasks finish.
8	Action: Send Email	Notifies the requester of the outcome.
9	Flow Logic: Else (Rejected)	Sets the item to Closed Incomplete and informs the requester.

Database: ServiceNow Tables

ServiceNow stores all data in its built-in relational database. The flow reads from and writes to the following tables:

Table	Label	Role in the Project
sc_cat_item	Catalog Item	Defines the Standard Laptop item and its variables
sc_request	Request	Parent record (REQ) created on order
sc_req_item	Requested Item	Child record (RITM) that triggers the flow
sc_task	Catalog Task	Work tasks for Procurement and IT Support
sysapproval_approver	Approval	Stores the manager's approval decision
sys_user / sys_user_group	User / Group	Requesters, managers and fulfilment groups
alm_hardware (optional)	Hardware Asset	Records the laptop once received

4. Setup Instructions

Prerequisites

- **ServiceNow instance:** a free Personal Developer Instance (PDI) from developer.servicenow.com.
- **Roles:** admin role on the instance.
- **Plugins:** Flow Designer and Service Catalog (active by default on a PDI).
- **Browser:** latest version of Chrome, Edge or Firefox.

Installation

1. Sign up at developer.servicenow.com and request a Personal Developer Instance.
2. Log in to the instance as admin and create a new Update Set (for example, "Laptop Procurement Automation") and set it as current.
3. Create the test users and groups: a requester, a manager, the Procurement group and the IT Support group. Assign the manager to the requester's Manager field.
4. Navigate to Service Catalog > Catalog Definition > Maintain Items and create the Standard Laptop catalog item with the variables listed in Section 3.
5. Publish the item to a catalog category (for example, Hardware) so it appears in the portal.
6. Open Flow Designer (All > Process Automation > Flow Designer) and create a new flow named "Standard Laptop Request".
7. Configure the trigger, actions and flow logic as described in Section 3, then click Save.
8. Click Activate to enable the flow. It is now ready to run.

5. Application Structure

ServiceNow is a platform-based application, so instead of client and server folders the solution is organized as configuration records. The table below maps each component to its location on the instance.

Layer	Component	Location in ServiceNow
Client	Catalog item and variables	Service Catalog > Catalog Definition > Maintain Items
Client	Service Portal page	Service Portal > Catalogs / Employee Center
Server	Flow and actions	Process Automation > Flow Designer
Server	Approval rules	Ask For Approval action inside the flow
Server	Notifications	System Notification > Email > Notifications
Data	Users, groups, roles	User Administration
Data	Tables and records	sc_request, sc_req_item, sc_task
Deployment	Update Set	System Update Sets > Local Update Sets

6. Running the Application

No separate servers need to be started, because the instance hosts both the portal and the flow engine. Follow these steps to run the application:

1. Log in to the instance and impersonate the requester user (User menu > Impersonate User).
2. Open the Service Portal at /sp and search for “Standard Laptop”.
3. Fill in the request form and click Order Now.
4. Impersonate the manager and approve the request from My Approvals.
5. Impersonate a member of the Procurement group and complete the catalog task, then repeat for the IT Support group.
6. Open Flow Designer > Executions to view the run of the flow and confirm that each action completed.

7. API Documentation

ServiceNow exposes the request data through REST APIs. The main endpoints used with this project are listed below. Replace <instance> with your instance name.

Method	Endpoint	Description
POST	/api/sn_sc/servicecatalog/items/{sys_id}/order_now	Orders the Standard Laptop item
GET	/api/now/table/sc_req_item	Lists requested items
GET	/api/now/table/sc_task	Lists catalog tasks
GET	/api/now/table/sc_request/{sys_id}	Fetches a single request

Example: Order the Laptop

Request: POST https://<instance>.service-now.com/api/sn_sc/servicecatalog/items/{sys_id}/order_now

```
{
  "sysparm_quantity": "1",
  "variables": {
    "requested_for": "<user_sys_id>",
    "business_justification": "New joiner laptop",
    "needed_by": "2026-10-15"
  }
}
```

Response (200 OK):

```
{
  "result": {
    "sys_id": "a1b2c3d4e5f60718293a4b5c6d7e8f90",
    "number": "REQ0010001",
    "request_number": "REQ0010001",
    "table": "sc_request"
  }
}
```

8. Authentication and Authorization

Authentication

- **User login:** users sign in to the instance with their credentials. The platform maintains a session and issues a session token.
- **REST API access:** Basic Authentication or OAuth 2.0 (bearer access tokens) can be used for the endpoints in Section 7.
- **Impersonation:** administrators can impersonate test users to validate each role without sharing passwords.

Authorization

Access is controlled through roles, groups and Access Control Lists (ACLs).

Actor	Role / Group	Permissions
Employee	itil / catalog user	Order the laptop and view own requests
Manager	Approver (manager of requester)	Approve or reject the request
Procurement Team	Procurement group	Work on the procurement catalog task
IT Support	IT Support group	Work on the set-up and delivery task
Administrator	admin	Configure catalog, flow and notifications

9. User Interface

The interface consists of the Service Portal catalog page, the request form and the approval and task views. Add the screenshots below.

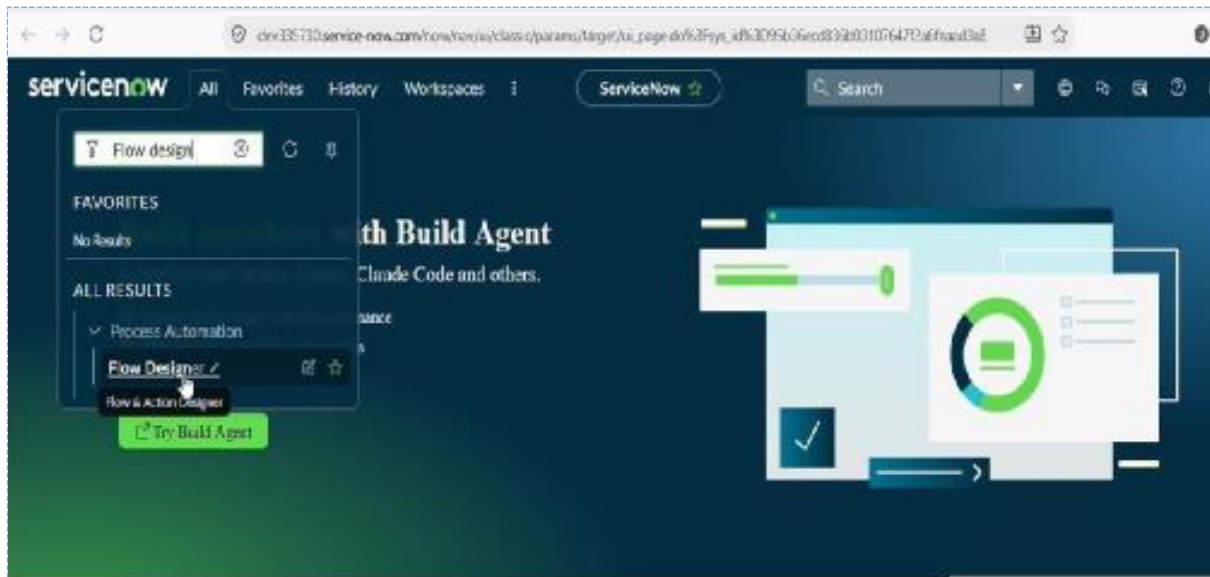


Figure 1: Service Portal home page showing the Standard Laptop item

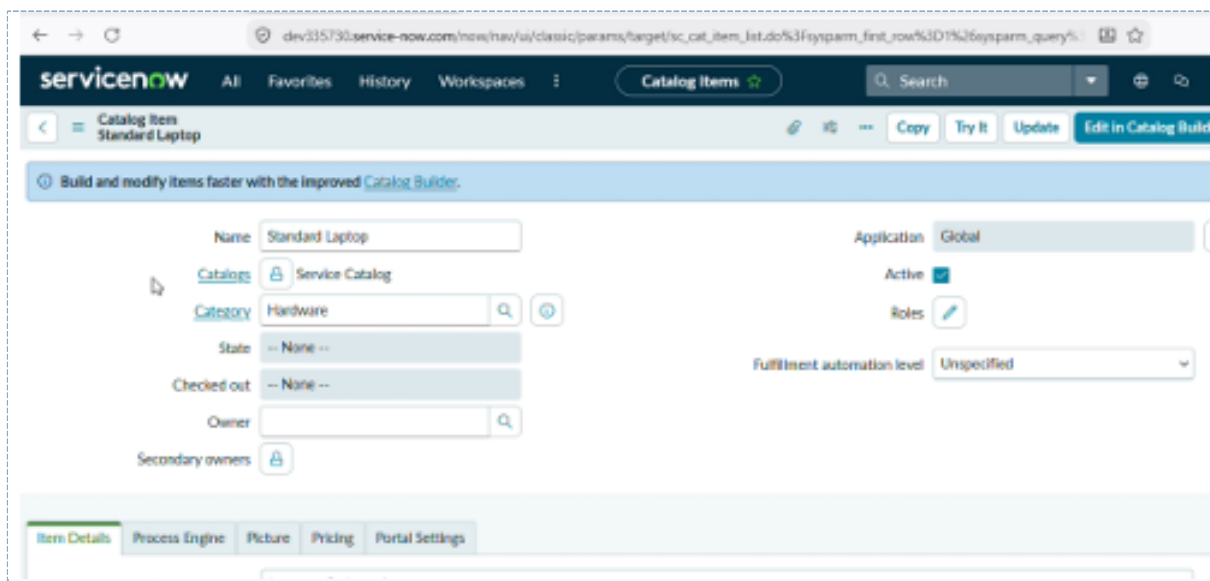


Figure 2: Standard Laptop request form with variables

10. Testing

Strategy

Testing was carried out manually using Flow Designer's built-in Test option and end-to-end runs by impersonating each user. Flow Executions were reviewed to confirm that every action completed successfully.

Tools Used

- Flow Designer Test feature for running the flow with sample inputs.
- Flow Designer > Executions for step-by-step debugging.
- Impersonate User to verify role-based access.

Test Cases

ID	Scenario	Expected Result	Status
TC01	Submit a valid laptop request	REQ and RITM are created and the flow starts	Pass
TC02	Manager approves	Procurement and IT tasks are created	Pass
TC03	Manager rejects	RITM is closed incomplete and the requester is notified	Pass
TC04	Both tasks completed	RITM moves to Closed Complete	Pass
TC05	Requester has no manager	Flow reports an error in Executions	Pass

11. Screenshots or Demo

The screenshot shows the ServiceNow interface for configuring a 'Standard Laptop' catalog item. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and a 'Catalog Items' dropdown. The main header displays the item name 'Standard Laptop' and action buttons: 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A blue banner below the header reads: 'Build and modify items faster with the improved Catalog Builder.' The configuration form is divided into two columns. The left column contains fields for 'Name' (Standard Laptop), 'Catalogs' (Service Catalog), 'Category' (Hardware), 'State' (None), 'Checked out' (None), 'Owner', and 'Secondary owners'. The right column contains 'Application' (Global), 'Active' (checked), 'Roles', and 'Fulfillment automation level' (Unspecified). At the bottom, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'.

Figure 3: Flow Designer canvas showing the complete flow

This screenshot shows the 'Process Engine' tab for the 'Standard Laptop' catalog item. The top navigation bar is similar to the previous screenshot, but the 'Catalog Items' dropdown is now 'Catalog Item - Standard Laptop'. The main header shows the item name and action buttons. Below the header, there is a section for 'Secondary owners'. The 'Process Engine' tab is active, displaying a blue banner that says: 'Select the appropriate process engine for the catalog item. Only one engine can be selected.' Below this banner are three input fields: 'Flow' (Standard laptop tasks), 'Workflow', and 'Execution Plan'. At the bottom, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. A 'Related Links' section at the bottom left contains links for 'Item Diagnostic', 'Show VA render type', and 'Run Point Scan'.

Figure 4: Flow execution details with all actions completed

12. Known Issues

- A request from a user without a manager assigned will fail at the approval step.
- The Personal Developer Instance hibernates after inactivity and may need to be woken up before use.
- The flow currently supports one standard laptop model only.
- Email notifications depend on the instance's email configuration, and a PDI does not deliver real emails by default.

13. Future Enhancements

- Add a stock check against the inventory so that available laptops are issued from stock before a purchase is raised.
- Support multiple laptop models and configurations through catalog variable choices.
- Integrate with a vendor or purchasing system for automatic purchase orders.
- Automatically create a hardware asset record and assign it to the user on delivery.
- Add SLA tracking and escalation for delayed approvals or tasks.
- Build a dashboard with Performance Analytics to report on procurement turnaround time.